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Coupled Heat Transfer Analysis in Regeneratively Cooled Thrust Chambers

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The coupled hot gas–wall–coolant environment that occurs in regeneratively cooled liquid rocket engines is studied by a computational procedure able to provide a quick and reliable prediction of thrust chamber wall temperature and heat flux as well as coolant flow characteristics, like pressure drop and temperature gain in the regenerative circuit. The coupled analysis is performed by means of a computational fluid dynamics solver of the Reynolds-averaged Navier–Stokes equations for the hot gas flow and by a simplified quasi-two-dimensional approach, which widely relies on semiempirical relations, for the coolant flow and wall structure heat transfer in the cooling channels. Coupled computations of the space shuttle main engine main combustion chamber are performed and compared with available literature data. Results show a reasonable agreement in terms of coolant pressure drop and temperature gain with nominal data, whereas the computed wall temperature peak is closer to hot-firing test data than to the nominal value.

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Nomenclature

b	=	channel width
$\mathcal{C}_{()}$	=	Nusselt number correction factors
D_h	=	channel hydraulic diameter
f_w	=	skin friction coefficient
h	=	channel height
h_w	=	heat transfer coefficient
k	=	thermal conductivity
$\dot{m}$	=	mass flow rate
Nu	=	Nusselt number
O/F	=	oxidizer-to-fuel mixture ratio
Pr	=	Prandtl number
p	=	pressure
q	=	heat flux
R_c	=	channel radius of curvature
R_h	=	channel hydraulic radius
Re	=	Reynolds number
r	=	engine radius
s	=	channel-axis abscissa
s_w	=	internal wall thickness
T	=	temperature
t_w	=	rib thickness
u	=	axial velocity component
x	=	engine-axis abscissa
y^+	=	dimensionless wall distance
ε	=	wall roughness
ξ	=	rough-to-smooth friction ratio
ρ	=	density

Subscripts

AW	=	adiabatic wall
c	=	chamber
f	=	film
HG	=	hot gas side
t	=	turbulent
w	=	wall
0	=	total

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I. Introduction

IN HIGH-PERFORMANCE liquid rocket engine thrust chambers an intense cooling is necessary to keep the wall temperatures within their allowed limits. A possible option is that of regenerative cooling, which is achieved by flowing a coolant into suitable channels surrounding the thrust chamber. Many studies have been published in the literature to solve the coupled heat transfer problem of hot gas, chamber wall, and coolant in thrust chambers. The most common approaches rely on the one-dimensional (1-D) analysis with Nusselt-type empirical relations for both hot gas and coolant flows. They have been extensively employed in thrust chamber cooling system design [1] and are still used [2] because of their low computational cost, even if their applicability is strictly limited to the operative ranges of the experimental data for which they are tuned. With a reasonable increase of complexity and computational cost, three-dimensional (3-D) heat conduction through the wall has been modeled in [3,4], while maintaining approaches based on semiempirical correlations for the coolant and the hot gas flows. On the other hand, to overcome the simplification arising from 1-D modeling of hot gas expansion, efforts have been made to describe the hot gas flow by means of multidimensional computational fluid dynamics (CFD) solvers [5], whereas the coolant flow and the wall are modeled by means of semiempirical methods. As a compromise between increasing modeling details and low computational cost, different model details have been adopted for each single problem as CFD for the hot gas flow, 3-D heat conduction model for the wall and 1-D semiempirical model for the coolant flow [6–9]. Recently, as the computational resources increase, coupled CFD analyses for the whole regeneratively cooled thrust chamber have been also performed [10–12].

The present study is a follow-on of that reported in [13]. The former study, aiming to validate accurate 3-D modeling of coolant flow and solid-material heat conduction, addressed the cooling circuit of the space shuttle main engine (SSME) main combustion chamber (MCC) considering the hot gas side as a boundary condition for the cooling circuit analysis. In particular, the hot gas side data gained from calorimetric thrust chamber test cases and relevant to the heat transfer coefficient and adiabatic wall temperature were considered. The lack of a detailed CFD description of the hot gas flowfield was recognized to be a major shortcoming, therefore a

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further study is carried out to present and discuss merits and limitations of a computational procedure able to describe the coupled hot gas–wall–coolant environment that occurs in most liquid rocket engines, with emphasis given to an accurate description of the hot gas flowfield. The result of the present approach provides a quick and reliable prediction of thrust chamber wall temperature and heat flux as well as coolant flow characteristics, like pressure drop and temperature gain in the regenerative circuit. In the present coupled approach, the hot gas side flowfield and heat transfer are computed by means of a CFD solver [14,15], whereas the coolant flow and the wall structure heat transfer are evaluated by means of a simplified quasi-two-dimensional (2-D) approach [16], which widely relies on semiempirical relations. The quasi-2-D model is able to account for the real fluid behavior of the coolant and the thermal stratification inside the channels. Coupled computations of the SSME MCC are carried out, analyzed in detail, and compared with available literature data.

II. Modeling

A comprehensive thermal model for a regeneratively cooled thrust chamber must account for convection from hot gas, conduction within the wall, and convection to the coolant. In its simplest form, regenerative cooling can be modeled as a steady heat transfer from a hot gas to a cold fluid, through a solid wall. This problem can be divided into three subproblems, which are defined as follows: 1) the hot gas flow, which is the turbulent flow of a mixture of hot gases in a thrust chamber; 2) the wall heat conduction, which is the heat conduction through the wall of the thrust chamber between the hot gas and the coolant; and 3) the coolant flow, which is the turbulent flow of the coolant in the channels surrounding the thrust chamber. These subproblems are coupled by the two steady-state balances of three heat fluxes: from hot gas to wall, through the wall, and from wall to coolant.

A. Hot Gas Flow

A Reynolds-averaged Navier–Stokes (RANS) approach is used to obtain CFD solutions of the hot gas flowfield and heat transfer in the thrust chamber. The steady-state numerical solutions are carried out by means of an in-house 3-D multiblock finite volume RANS equations solver, second-order accurate in space [14], able to treat multicomponent mixtures of thermally perfect gases. Turbulence is described by means of the Spalart–Allmaras one-equation model [17], a constant turbulent Schmidt number is adopted to model turbulent diffusivity, and a constant turbulent Prandtl number is adopted to model turbulent conductivity. The present solver has been validated in different operative conditions against experimental data [15,18,19].

Frozen flow assumption is adopted without aiming to describe the details of injection and combustion of propellants in the thrust chamber. The chemical composition of the frozen mixture is evaluated by means of the Chemical Equilibrium and Applications (CEA) program [20] at the chamber pressure and oxidizer-to-fuel mixture ratio. The mixture of hot combustion products enters the computational domain through the whole thrust chamber cross section (i.e., "full-inlet" assumption [21]), and its chemical composition is kept constant through the chamber. The evaluated adiabatic flame temperature is enforced as the chamber stagnation temperature.

Neglecting injection and combustion details is a reasonable simplification for the goals of the present study. In fact, the injected hot gas flow enthalpy matches the hot gas energetic content although neglecting the complex phenomena related to the propellant injection confined to the near injector plate region. Besides that, the present evaluation of heat transfer does not take into account the effect of mixture ratio shift and flow direction biasing that is often made at the peripheral injectors. In fact, lower values of O/F (more fuel) and flow direction away from the wall [22] may be used as a mean to reduce thermal loads on the chamber wall, at least in the region closer to the injection plate. Nevertheless, in full-scale engines and sufficiently downstream of the injector plate, the full-inlet assumption is in agreement with the results obtained with more complex approaches, which account for the phenomena involved in the propellant injection and combustion [23]. Hence, being aware of the lack of modeling of the near injector plate phenomena, in the present coupled analysis the full-inlet assumption is made as a good compromise between accuracy in the throat region and low computational cost.

B. Coolant Flow and Wall Heat Conduction

A simplified quasi-2-D approach, presented and validated against experimental data in [16], is used to study the coupled problem of coolant flow and wall structure heat transfer in rectangular cooling channels. In Fig. 1, a typical regenerative cooling circuit arrangement is shown focusing on the rectangular cooling channel dimensions.

The cooling channel flow model is developed by using the steady-state conservation laws of mass, momentum, and energy, taking into account the effects of heat transfer and friction. In particular, the mass and momentum governing equations are written in a 1-D form, whereas a simplified arrangement of the 2-D energy equation is considered. In fact, the fluid energy equation is reduced to a 2-D balance in the streamwise and radial direction, and, as a consequence, this flow model is referred to as "quasi-2-D." Moreover, to accurately describe the wall heat flux, also the axial component of fluid velocity u and wall temperature T_w are assumed to vary along the radial direction. As in 1-D analysis, the transverse velocity components are neglected because they are much smaller than the axial component. According to the preceding hypotheses, the fluid pressure only depends on streamwise coordinate position x, whereas the fluid temperature and axial velocity and wall temperature are left to vary also in the radial direction. The model, aiming to study any fluid evolving through cooling channels, considers a generic equation of state, and thus the coolant fluid can be either a compressible gas or a supercritical fluid or a liquid. The turbulent thermal conductivity, fluid skin friction, and coolant wall heat transfer coefficients are evaluated by semiempirical relations provided in the literature. In this study, turbulent thermal conductivity k_t in the radial direction is modeled according to [24]

$$\frac{k_t}{k} = 0.008 Re^{0.9} \tag{1}$$

where k is the fluid thermal conductivity and Re is the Reynolds number. The coolant friction is modeled by means of Colebrook's correlation [25], because of its capability to evaluate the rough wall effect:

$$\frac{1}{\sqrt{f_w^0}} = -2 \log\left(0.27\frac{\varepsilon}{D_h} + \frac{2.51}{Re\sqrt{f_w^0}}\right) \tag{2}$$

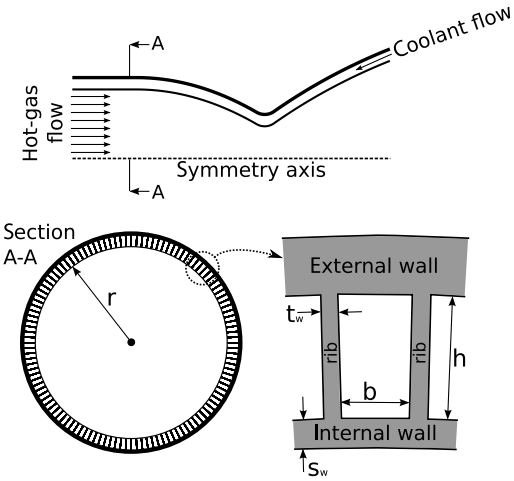

Fig. 1 Schematic of regenerative cooling circuit arrangement with rectangular cooling channels.

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where ε is the wall roughness and D_h is the channel hydraulic diameter. The coolant heat transfer coefficient is modeled according to Niino's correlation [26], because it is fitted to supercritical hydrogen flow in rough curved channels with strong wall coolant temperature difference and high Reynolds number, as in the case under investigation. Niino's semiempirical relation is generally expressed as

$$Nu = Nu^0 \mathcal{C}_\xi \mathcal{C}_x \mathcal{C}_I \tag{3}$$

where Nu^0 is the Nusselt number for a nonadiabatic smooth tube and is a function of the Reynolds number and the Prandtl number. To take into account the effect of variable temperature, these nondimensional parameters are evaluated at the film temperature, which is the mean value between the fluid and the wall temperature:

$$Nu^0 = 0.062 Re_f^{0.7} Pr_f^{0.4} \tag{4}$$

The term $\mathcal{C}_\xi$ is a roughness factor to correct for the surface roughness, $\mathcal{C}_x$ is an entrance factor to correct for the heat transfer coefficient in the thermally undeveloped region, and $\mathcal{C}_I$ is a curvature factor to account for a secondary flow effect induced in a curved channel:

$$\mathcal{C}_\xi = \frac{1 + 1.5 Pr^{-1/6} Re^{-1/8}(Pr - 1)}{1 + 1.5 Pr^{-1/6} Re^{-1/8}(Pr\xi - 1)}\xi \tag{5}$$

$$\mathcal{C}_x = 1 + \left(\frac{s}{D_h}\right)^{-0.7}\left(\frac{T_w}{T}\right)^{0.1} \tag{6}$$

and

$$\mathcal{C}_I = \left[Re\left(\frac{R_h}{R_c}\right)^2\right]^{\pm 0.05} \tag{7}$$

where $\xi = f_w^0(Re, \varepsilon)/f_w^0(Re, \varepsilon = 0)$ is the ratio of the rough tube isothermal friction factor to the smooth one, s is the channel-axis abscissa whose origin is the channel inlet, R_h is the hydraulic radius of the channel cross section ($R_h = D_h/4$), R_c is the channel radius of curvature, and the sign $+$ or $-$ in the exponent of the expression for $\mathcal{C}_I$ denotes the concave or convex curvature, respectively.

C. Coupling: Hot Gas Flow and Wall Heat Conduction

The quasi-2-D model is able to solve the coupled wall and coolant flow evolution once thermal boundary conditions are provided. In particular, the model, suited to rectangular channels as those shown in Fig. 1, requires thermal boundary conditions at the external and internal walls. The thermal boundary condition can be one of the following: assigned wall temperature, assigned heat flux, or assigned convective heat transfer coefficient and adiabatic wall temperature. On the outer side of the rectangular channel an adiabatic condition is assumed, whereas the hot gas side thermal boundary condition is obtained by the CFD solver. Of course, in principle, neither temperature nor wall heat flux are known, because they depend on the thermal equilibrium between flows and wall. A first solution can be computed, for instance, enforcing a first tentative distribution of wall temperature to the CFD solver, then passing the resulting heat flux to the quasi-2-D solver, comparing the wall temperature computed by the quasi-2-D model with the first given value. Subsequently, the CFD computation enforcing the wall temperature computed by the quasi-2-D solver can be repeated iteratively until convergence is reached. Alternatively, the iterative procedure could be done in the same manner by enforcing a first tentative heat flux distribution. Although these processes could successfully converge, a more efficient way is that of enforcing convective heat transfer coefficient and adiabatic wall temperature as a boundary condition of the quasi-2-D solver. In fact, in this case, only the heat transfer coefficient has to be adjusted, which only weakly depends on wall temperature. Therefore, in the present coupled analysis the convective heat transfer coefficient approach is used to provide the boundary condition to the quasi-2-D solver. In particular, the coupling procedure is described by the following steps:

1) A first CFD simulation of the hot gas flow is made assuming null heat flux at wall. The useful output of this simulation is the wall temperature that, as is obtained assuming zero wall heat flux, is the adiabatic wall temperature T_{AW}, which is used subsequently to evaluate the convective heat transfer coefficient.

2) CFD computation of the hot gas flow enforces the wall temperature ($T_{w,\mathrm{HG}} = 700$ K at first iteration, the result provided by quasi-2-D computation for the next iterations) as a boundary condition, heat transfer coefficient $h_{w,\mathrm{HG}}$ is evaluated from computed heat flux $q_{w,\mathrm{HG}}$, enforced wall temperature $T_{w,\mathrm{HG}}$, and adiabatic wall temperature T_{AW} evaluated in step 1:

$$h_{w,\mathrm{HG}} = \frac{q_{w,\mathrm{HG}}}{T_{\mathrm{AW}} - T_{w,\mathrm{HG}}} \tag{8}$$

3) Quasi-2-D computation of the coupled coolant flow and wall thermal evolution uses $h_{w,\mathrm{HG}}$ values from step 2 and T_{AW} evaluated in step 1, and the coupled coolant flow and wall analysis provides a new value of $T_{w,\mathrm{HG}}$.

Iterations are made repeating steps 2 and 3. The process ends when the difference between the computed values of $T_{w,\mathrm{HG}}$ in two consecutive global iterations is smaller than an assigned tolerance. The first global iteration is able to provide a very good evaluation of the heat transfer coefficient, which is only slightly refined by the following coupling steps, as will be shown for the test case of interest in Sec. IV.B.

III. Test Case

For the sake of comparison with experimental data, the coupled hot gas–wall–coolant analysis is carried out for the SSME MCC, whose data relevant to the regenerative cooling circuit and the thrust chamber profile have been published in the open literature [9,22,27,28]. The SSME is an oxygen–hydrogen staged-combustion engine that uses the fuel as coolant and presents a throttle range between 67 and 109% of rated thrust. In particular, the selected data refer to the standard throat version of the MCC, whose throat diameter is 261.75 mm with a contraction ratio of three and an expansion ratio of five. The liner is made of NARloy-Z copper alloy and provides the coolant flowpath for the MCC. It has 390 milled axial coolant channels that are closed out by an electroforming process, which deposits a copper barrier, followed by a nickel Inconel 718 structural closeout over the coolant channels [9,27]. Coolant moves from the inlet manifold, located at the MCC nozzle exit, to the exit manifold, located at the injector-face section. The channel geometry properties, defined in Fig. 1 and taken from [27], are reported in Fig. 2.

Among the available data, the most complete set for the standard throat SSME-MCC refers to the full-power-level operative condition (FPL, 109% of rated thrust) [9,27], with chamber pressure $p_c = 225.87$ bar. The injector plate is composed of 600 coaxial injectors arranged in 13 rings providing the overall oxidizer-to-fuel mixture ratio $O/F = 6.0$. The outer injector ring is operated at a reduced mixture ratio to reduce the temperature of the combustion products near the wall, and the oxidizer post is biased directing the flow away from the combustion chamber wall [22]. The coolant operative conditions at FPL are taken from [9]: overall coolant mass flow rate is $\dot{m} = 14.306$ kg/s (i.e., 36.68 g/s for a single channel), inlet temperature is 53.89 K, and inlet pressure is 445.47 bar. Coolant thermodynamic and transport properties used in the fluid solver are taken from [29], whereas the solid-material thermal conductivity is taken from [30] and refers to that of NARloy-Z copper alloy at 533 K: $k_w = 316$ W/(m · K).

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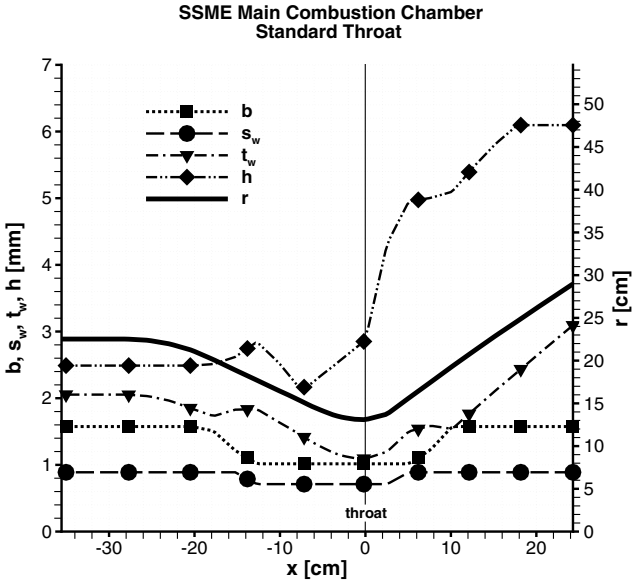


Fig. 2 SSME MCC geometry (data taken from [27]).

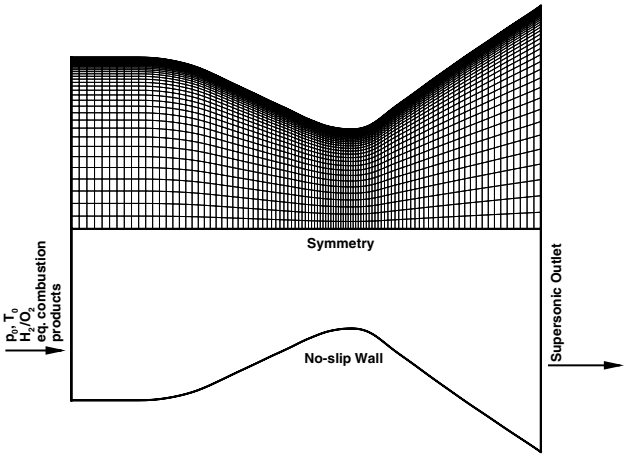


Fig. 3 Numerical grid (medium grid size) (top) and boundary conditions (bottom) for the hot gas side domain.

IV. Results

In this section, the uncoupled analysis of the hot gas flow heat transfer is discussed first to show the capability of the CFD approach to provide a reliable hot gas side heat transfer coefficient and to quantify the numerical error relevant to the hot gas flow results. Then, the hot gas–wall–coolant coupled analysis is discussed and the numerical results are compared with experimental data and numerical evaluations retrieved in open literature.

A. Uncoupled Analysis

The aim of the preliminary uncoupled analysis is the evaluation of the hot gas side heat transfer coefficient and its comparison against literature data. In particular, the hot gas side heat transfer coefficient is computed from Eq. (8), where the wall heat flux results from a CFD computation carried out with an isothermal wall boundary condition ($T_{w,\mathrm{HG}} = 700$ K), and the adiabatic wall temperature results from a CFD computation obtained enforcing an adiabatic wall.

The hot gas flow domain is discretized by means of the structured 2-D axis-symmetric grid shown in Fig. 3. Volumes are clustered toward the wall to properly solve the boundary layer up to the viscous sublayer. At the subsonic inlet boundary, chamber stagnation pressure and temperature are prescribed together with velocity direction. The outlet section is set as supersonic outflow (i.e., no prescribed conditions), the lower boundary is an axis of symmetry, and the upper boundary is a no-slip wall, as shown in Fig. 3.

A grid convergence analysis is made in the case of the isothermal wall to verify that solutions are independent of the selected grid. In Table 1, the number of volumes of the three grids adopted for the analysis are summarized together with the minimum volume dimension at the wall. The y^+ value obtained with the three grid levels along the wall is plotted in Fig. 4a. The numerical solutions obtained with the three grid levels are shown in Fig. 4b in terms of wall heat flux. These solutions are used to estimate the exact solution by Richardson extrapolation [31], which is used to evaluate the numerical error. The numerical error is thus defined as the difference between the Richardson extrapolated solution and the solution obtained with the selected grid. The numerical error in terms of wall heat flux along the nozzle is shown for the three grid levels together with the Richardson extrapolated solution in Fig. 4c.

Note that the medium grid size is able to provide a solution with a numerical error lower than 2% and a y^+ value of the order of unity all along the thrust chamber. Hence, the solution obtained by means of the medium grid size is adopted in the following for the heat transfer analysis.

The hot gas side convective coefficient evaluated by means of the CFD numerical simulation is compared in Fig. 5a with that measured in the calorimetric test case and scaled as described in [28] and to that predicted by means of the semiempirical Bartz's equation [32].

The convective heat transfer coefficient shown in Fig. 5a is evaluated from Eq. (8) assuming the adiabatic wall temperature plotted in Fig. 5b for each method. For the CFD evaluation, the adiabatic wall temperature along the thrust chamber is evaluated by means of a numerical simulation where the adiabatic wall is enforced as a boundary condition assuming a frozen composition once the hot gas mixture is injected in the chamber. The scaled calorimetric evaluation is related to T_{AW} evaluated by means of the CEA program assuming chemical equilibrium all over the chamber as reported in [28]. Finally, in the Bartz's equation, T_{AW} is evaluated by means of the CEA program under the Bray hypothesis, for which the mixture composition is evaluated at the chemical equilibrium up to the throat section, and then it is considered frozen [33]. The three approaches provide similar T_{AW} up to the convergent part of the nozzle and then follow different trends. This is related to the different approaches they use to evaluate the mixture composition inside the thrust chamber, which, in turn, affects the thermodynamic and transport properties of the flowing gas.

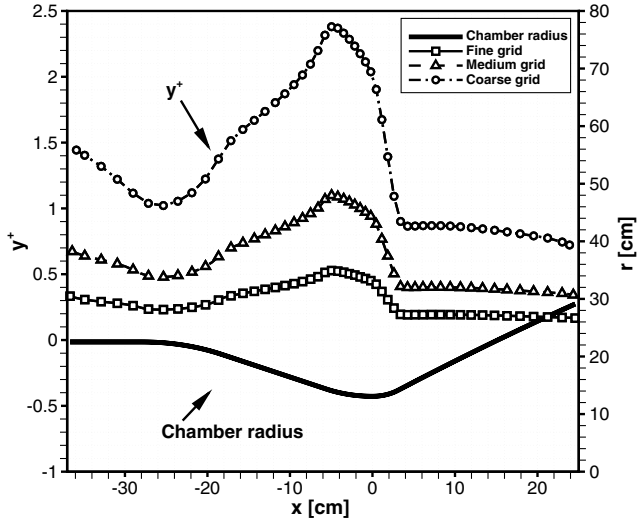
The CFD numerical solution for the hot gas side convective heat transfer is characterized by an expected decreasing trend in the first part of the chamber ($x < -25$ cm) related to the adopted near injector plate model (Sec. II.A) that does not follow the reference calorimetric-curve trend. Downstream, the coefficient increases, as expected, reaching its maximum value, 58 $\mathrm{kW/(m^2 \cdot K)}$), at $x = -4$ cm upstream of the throat section. Then the convective coefficient decreases with two different slopes: the steeper one across the throat section and the milder one downstream in the divergent section. The scaled calorimetric evaluation shows the same trend but different values: it is higher in the convergent part, whereas it is lower in the divergent part. The Bartz's formula prevision does not follow the same trend as the other two methods, showing its maximum value at the throat section and a milder decrease in the divergent part of the nozzle.

B. Coupled Analysis

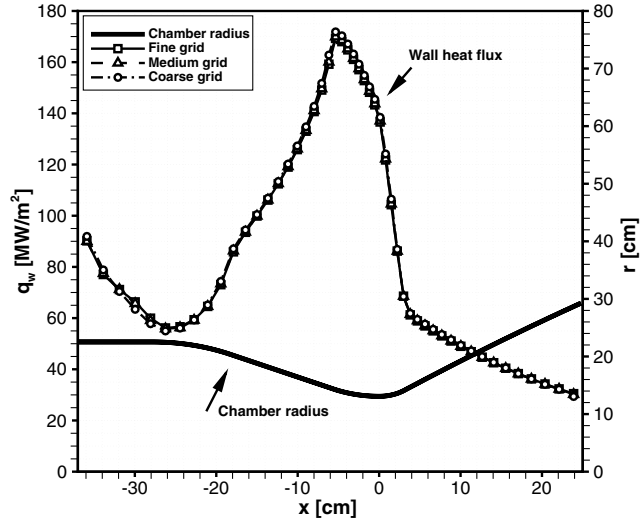
A typical behavior of the hot gas side heat transfer coefficient and wall temperature evaluated at different global iteration steps is

Table 1 Grid convergence analysis: volumes and minimum volume dimension

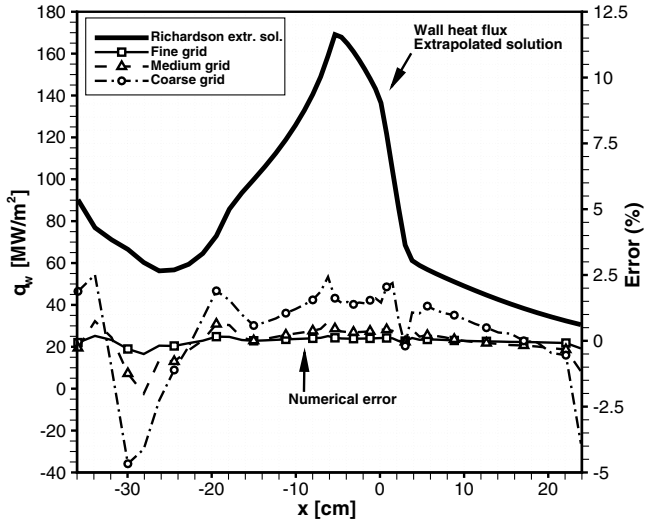
Grid refinement	Volumes (axial × radial)	$\Delta y_{\min}$
Coarse grid	50×45	2 μm
Medium grid	100×90	1 μm
Fine grid	200×180	0.5 μm



a) y^+ along the wall

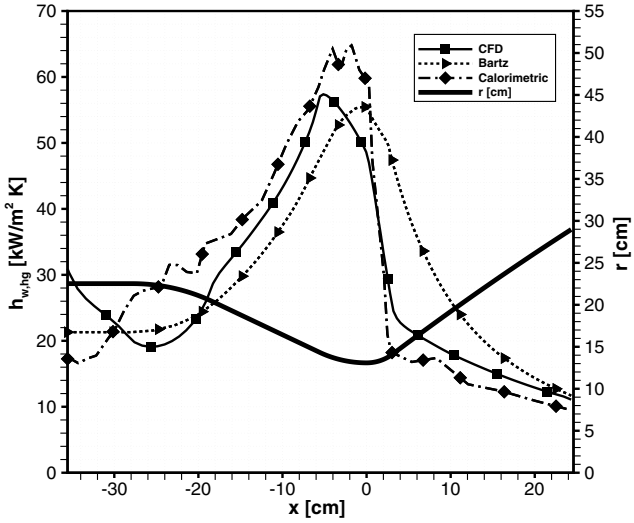


b) Wall heat flux along the chamber

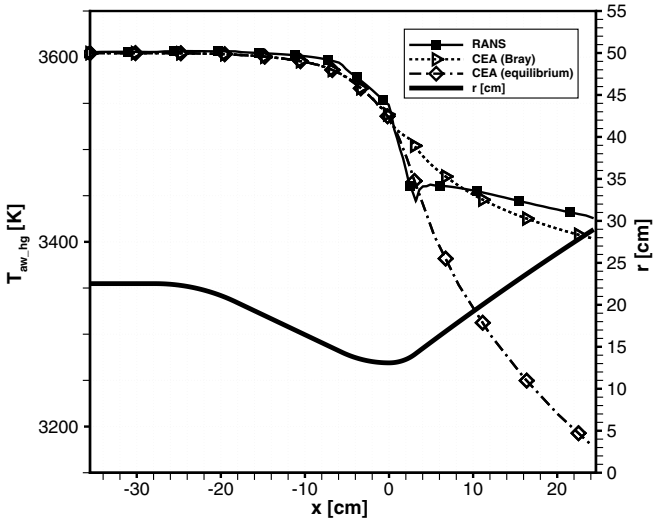


c) Richardson extrapolated solution and numerical error

Fig. 4 CFD solution grid independence analysis with three grid levels: isothermal test case ($T_{w,\mathrm{HG}} = 700$ K).



a) Hot gas side heat transfer coefficient



b) Adiabatic wall temperature

Fig. 5 Hot gas side coupling parameters.

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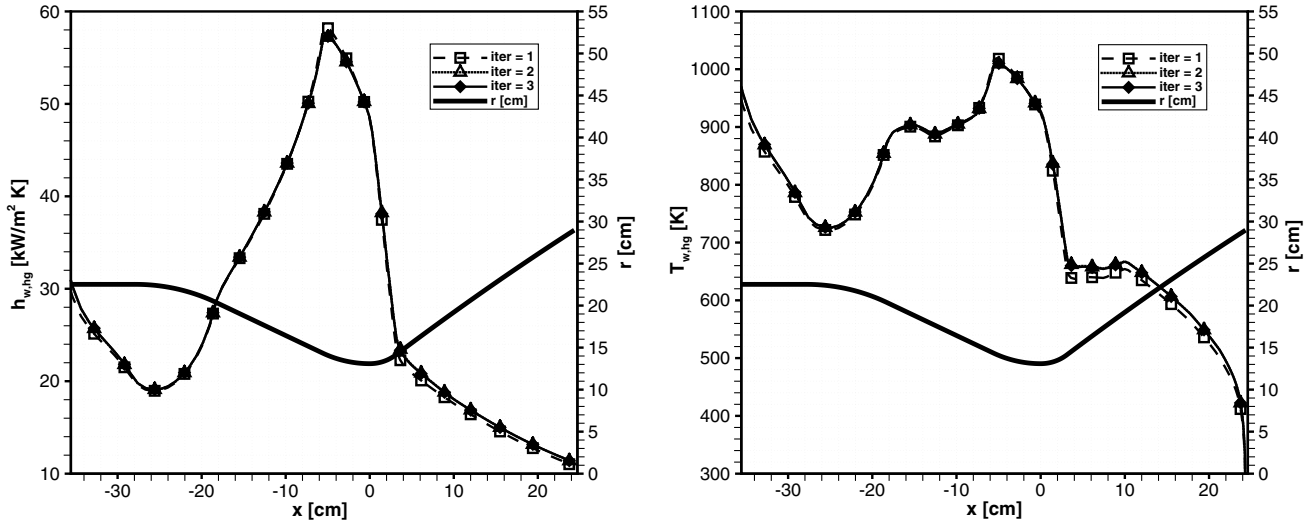

a) Hot-gas side heat transfer coefficient

b) Hot-gas side wall temperature

Fig. 6 Coupling process convergence history.

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presented in Fig. 6 for the test case of interest in this study. It can be noticed that, after the second global iteration step, the solution is very close to the previous one. This is clearly noticed in Fig. 6a for the hot gas side heat transfer coefficient, where the maximum difference, located at the abscissa $x = 5$ cm, between the first two iterations is 1.5% and vanishes after the second global iteration. Good convergence can also be noticed in Fig. 6b, where it is shown that wall temperature differences between the first two global iterations are lower than 15 K along most of the engine and with a maximum difference of 23 K located at the axial abscissa $x = 4$ cm. Also in this case, after the second global iteration, convergence is fully reached, as the solutions do not appreciably change anymore.

The results obtained with the present approach (referred to as "CFD + quasi-2-D") are compared with available literature data in Fig. 7. Besides estimations reported in [9] (referred to as "Wang and Luong"), which also include the heat fluxes predicted by Rocketdyne (referred to as "Design Method A"), the present solutions are obtained with two different levels of cooling channel wall roughness: 0 μm (smooth) and 0.23 μm (rough). The latter value of roughness has been selected in order to have the same pressure drop as [9].

Although the actual value of the surface roughness for the SSME cooling channels is not declared in the open literature, the typical value for a milled channel in copper and copper-base alloy chambers is between 0.1 and 0.5 μm [8,34,35]. Note that the selected value of surface roughness (0.23 μm) is within this range.

Pressure loss behavior is presented in Fig. 7a. The pressure loss estimated with the present approach and the hypothesis of the smooth wall is 67 bar, which is 21% lower than that of [9] (87.9 bar). Pressure loss discrepancy could be partially explained by the assumption of smooth walls. In fact, with the hypothesis of the cooling channel with roughness of 0.23 μm, the present coupled approach agrees with results of [9]. The coolant temperature gain is shown in Fig. 7b, where it can be seen that the selected level of roughness implies mild differences with respect to the smooth case and that the present approach provides similar results with respect to [9]. In fact, differences among curves are always within 10 K. According to the limits of the full-inlet approach adopted for the hot gas side modeling (Sec. II.A), the CFD solutions of the hot gas side heat flux (Fig. 7c) and wall temperature (Fig. 7d) do not include the first part of the numerical domain ($x < -26.5$ cm). The comparison of hot gas side heat fluxes (Fig. 7c) shows that the differences among the present results and those presented in [9] are of the order of 10%. In fact, the peak value, which is achieved in the throat region, is 161.5 MW/m^2 with design method A and 160.8 MW/m^2 with the Wang and Luong estimation, whereas it is 147.82 and 143.91 MW/m^2 in the present study with the hypothesis of the rough or smooth wall, respectively. In the other regions of the thrust chamber, the present results are within those presented in [9]. In particular, present results are close to the Wang and Luong estimations in the divergent part and close to design method A in the convergent part. Even larger differences can be seen analyzing the wall temperature behavior (Fig. 7d). In particular, the computed hot gas side wall temperature is much larger than expected, the maximum value, in the throat region, being 1071 or 1011 K for the present model with the hypothesis of the smooth or rough wall, respectively, compared with the value of 850 K declared in the open literature [9,30]. However, the qualitative behavior of $T_{w,\mathrm{HG}}$ is similar to that of *Wang and Luong*: a peak value slightly upstream of the throat, a local minimum at nearly $x = 13$ cm, and a strong variation downstream of the throat, followed by a local maximum. It is worth noticing that the beneficial effect of the wall roughness on the wall temperature implies a reduction of 60 K. Moreover, the computed hot gas side characterization is in line with that of [9]. In fact, the computed hot gas side heat transfer coefficient $h_{w,\mathrm{HG}}$ and adiabatic wall temperature T_{AW} are able to provide, via Eq. (8), the nominal heat flux and wall temperature in the throat region of the SSME MCC ($q_{w,\mathrm{HG}} = 160$ MW/m^2 and $T_{w,\mathrm{HG}} = 850$ K [9,27,30]). Therefore, the achieved wall temperature overshoot with respect to the result of [9] can be addressed to the coolant flow modeling only. Concerning this latter aspect it is worth noticing that the present coolant flow modeling provides results quite close to full-3-D CFD simulations, as shown recently in [13]. Moreover, experimental evidences based on full-scale hot-fired engines have shown that the operating temperature of the hot wall is much higher than expected, up to more than 1030 K [36,37]. This result is close to that achieved with the present numerical approach and gives a strong confidence in its validity and accuracy. Of course, the throat-region heat flux seen in Fig. 7c, which is lower than the nominal one [9,27] ($\sim$160 MW/m^2), is a direct consequence of the higher hot gas side wall temperature (Fig. 7d), for a given hot gas side heat transfer coefficient and adiabatic wall [Eq. (8)].

The hot gas side wall temperature obtained with the different heat transfer coefficients presented in Fig. 5a (CFD, Bartz, and calorimetric solutions) and with the hypothesis of the rough wall is presented in Fig. 8 together with that presented in [9]. Both the estimations achieved with Bartz's correlation and with the scaled heat transfer coefficient from the calorimetric chamber [28] give a maximum wall temperature in the throat region that is larger than that achieved with the CFD computation. In particular, the larger value of the wall temperature derived from the calorimetric chamber data is due to the larger value of $h_{w,\mathrm{HG}}$ (Fig. 5a); in this case the peak value in the throat region is 1084 K and the qualitative behavior of $T_{w,\mathrm{HG}}$ is similar to that estimated by Wang and Luong. Furthermore, the estimation achieved with Bartz's heat transfer coefficient has a peak value of 1057 K, which is located, differently from other predictions,

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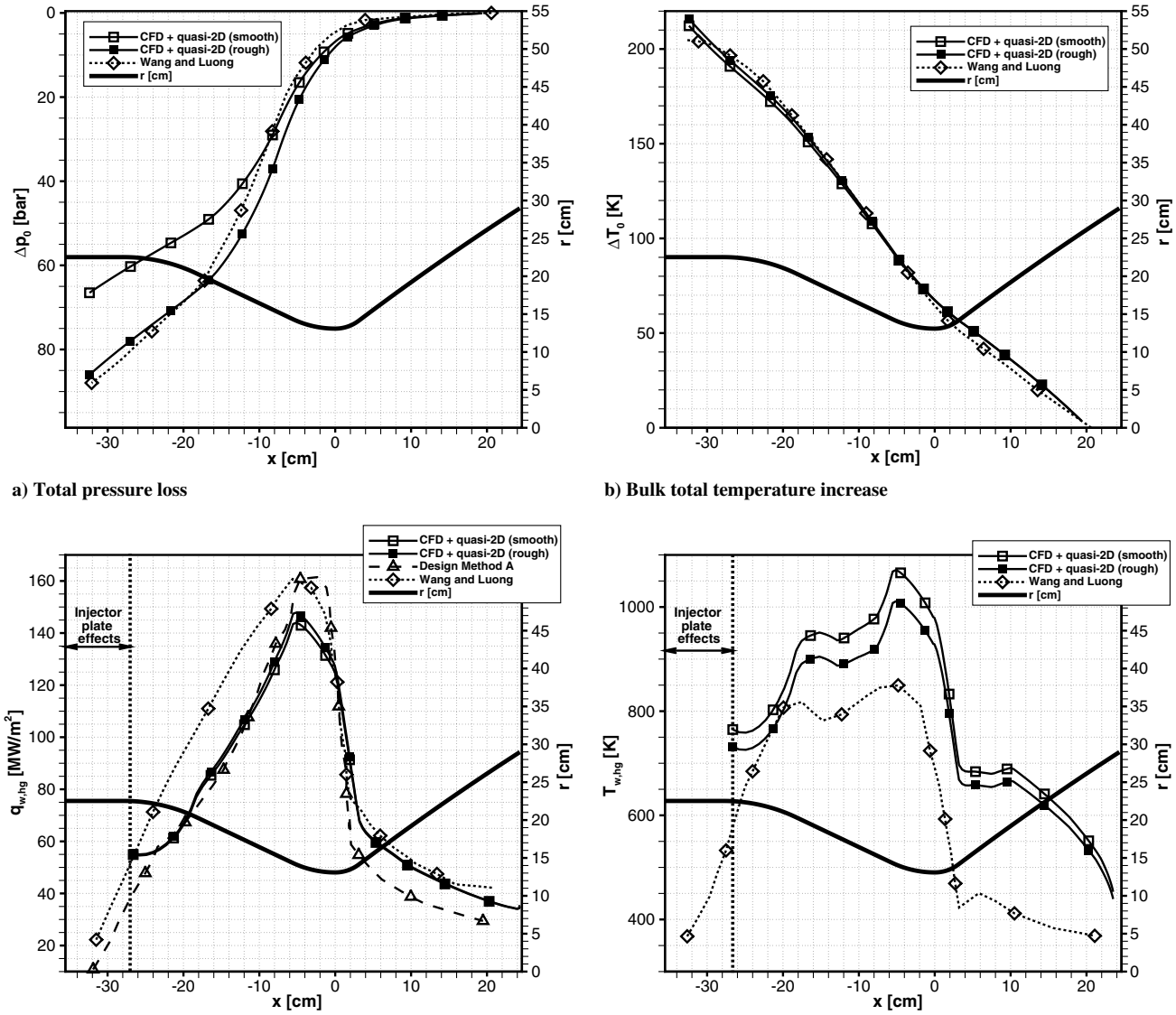

a) Total pressure loss

b) Bulk total temperature increase

c) Hot-gas side heat flux

d) Hot-gas side wall temperature

Fig. 7 SSME MCC, standard throat, FPL operative condition: comparison of present and literature numerical simulations.

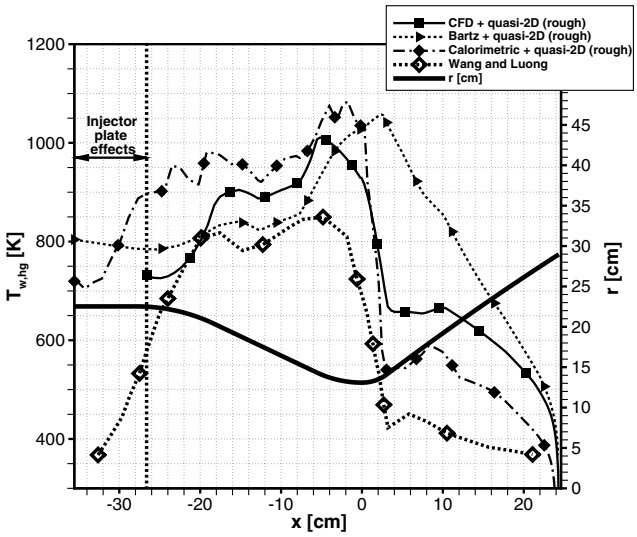

Fig. 8 SSME MCC, standard throat, FPL operative condition: comparison of present and literature numerical solution of hot gas side wall temperature.

downstream of the throat section. In fact, due to the strongly different behavior of $h_{w,\mathrm{HG}}$ predicted with Bartz's correlation with respect to the others (Fig. 5a), the wall temperature is far larger in the divergent part of the engine and its general behavior cannot be compared with that of [9].

V. Conclusions

A coupled procedure for the simulation of the thermal behavior of the hot gas–wall–coolant environment in a regeneratively cooled liquid rocket engine has been set up and demonstrated. The analysis of the SSME MCC has shown a reasonable agreement with published data of coolant pressure drop and temperature gain. Also, data concerning the hot gas side wall temperature are in agreement with experimental measures taken from full-scale hot-fired engines, even if they are in disagreement with the nominal design point. In particular, maximum wall temperature is higher than the nominal value. As a consequence, the maximum computed hot gas side heat flux results to be lower than the nominal value, but close to its measured values. Considering the hot gas side heat transfer coefficient evaluated by Bartz's model and the scaled data of a calorimetric chamber of the SSME MCC, the wall temperature is 100 K higher than that predicted with the present approach, and in the case of Bartz's model the general behavior of the wall temperature is in evident disagreement with the other solutions. In conclusion, the

present coupled approach, although based on simple modeling of the cooling circuit and engine thrust chamber, seems to be able to predict results in a very good agreement with experimental data.

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